

Why there is no closed formula for #DS: double-stranded reconstruction is #P-hard

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Abstract

Single-stranded reconstruction counts are given by one determinant, the BEST theorem. The double-stranded count $\#DS(S, k)$ of Bio 8 is a sum of such determinants over the points of a lattice in a box. Bio 12 and 13 evaluated it for $\phi X174$ by frontier elimination and by visiting $1.24 \cdot 10^9$ points. It is natural to ask for a formula that avoids the sum: a Gauss sum over a discriminant form, a Pfaffian, anything computable in polynomial time. We show that no such formula exists unless $\text{FP} = \#P$. From every connected loopless graph G of maximum degree 4 with even degrees, we build in polynomial time a circular DNA sequence S of length linear in $|E(G)|$ and an even k such that

$$\#DS(S, k) = 2 \varepsilon(G),$$

where $\varepsilon(G)$ is the number of Eulerian circuits of G . The number of Eulerian circuits is $\#P$ -complete for 4-regular graphs (Ge and Štefankovič, after Brightwell and Winkler). The mechanism is simple: palindromic k -mers make the reverse-complement involution act trivially on vertices, so that choosing a strand for each ρ -orbit is the same as orienting an edge. Two corollaries follow. The number of box points, Bio 12's first column, is the number of Eulerian orientations of G , which is $\#P$ -complete as well. And on this class the generation conjecture of Bio 17 holds, by Kotzig's theorem. All identities are verified exactly on 23 graphs by `bio19_verify.py`. The lattice sum, and the frontier and enumeration methods used to evaluate it, are therefore not an artefact of the approach; they are what the problem requires.

Keywords. computational complexity, $\#P$ -completeness, Eulerian circuits, Eulerian orientations, BEST theorem, de Bruijn graph, reverse complement, palindromes, genome assembly.

1 The question

For one strand the count of sequences with a given $(k + 1)$ -mer spectrum is a determinant times factorials [6]. For two strands, Bio 8 wrote

$$\#DS(S, k) = \sum_{f \in \Lambda^-, |f| \leq m, f \equiv m \pmod{2}} N_{\text{seq}}\left(\frac{m+f}{2}\right)$$

[2]. Bio 12 counted the 2 481 687 900 points of the box for $\phi X174$ at $k = 10$, and Bio 13 evaluated the sum, 31 925 753 246 212 [3, 4]. Each step needed an exponential-size object: a frontier of

width 26, or a visit to every point. A proposal for this paper asked for a “meshless” formula instead. It would be a Gauss sum built on the symmetric linking form of the double cover’s sandpile group [1, Rem. 5], or a Pfaffian in a Clifford algebra, evaluated in time polynomial in the number of vertices. We prove that such a formula cannot exist unless every problem in #P is solvable in polynomial time.

2 Eulerian circuits and their orientations

Let G be a connected loopless multigraph whose degrees are all even, with at least three edges. An *Eulerian circuit* is a cyclic sequence of its edges, traversing each once, considered up to rotation and reversal. Let $\varepsilon(G)$ be their number. An *Eulerian orientation* directs each edge so that every vertex has equal in- and out-degree. For a balanced digraph O with labelled arcs, $\text{ec}(O) = \tau(O) \prod_v (d_v^+ - 1)!$ counts its Eulerian circuits up to rotation [6].

Lemma 1.
$$\sum_{O \text{ Eulerian orientation of } G} \text{ec}(O) = 2 \varepsilon(G).$$

Proof. An Eulerian circuit of G , traversed in one of its two directions, orients every edge. It is balanced at every vertex, since each visit enters and leaves once, and it gives a directed Eulerian circuit of that orientation. Conversely, a directed Eulerian circuit of an orientation, forgetting the arrows, is a traversed Eulerian circuit of G . So the left side counts pairs (circuit, direction). The two directions of a circuit are different directed circuits, because they traverse the first edge oppositely. $\square$

Theorem 2 ([7, 8]). *Computing $\varepsilon(G)$ is #P-complete, even for 4-regular graphs.*

Counting the Eulerian orientations of a graph is also #P-complete [9].

3 Genomes whose double cover is a given graph

Let G have vertex set $\{1, \dots, n\}$, $m \geq 3$ edges, and all degrees in $\{2, 4\}$. Fix an even word length $k = 2h$ and a spacer length ℓ . The construction has three steps.

1. To each vertex x assign a palindrome $P_x = u_x \text{rc}(u_x)$ with $|u_x| = h$.
2. Compute an Eulerian circuit $x_0 \xrightarrow{e_1} x_1 \xrightarrow{e_2} \dots \xrightarrow{e_m} x_m = x_0$ of G . Spell

$$S = P_{x_0} \Sigma_1 P_{x_1} \Sigma_2 \dots P_{x_{m-1}} \Sigma_m \quad (\text{circular}),$$

with one spacer Σ_i of length ℓ per traversal.

3. Assign the boundary letters of the spacers. P_x occurs $\deg(x)$ times in S and $\text{rc } S$ together. The letters that follow it are the first letters of the outgoing spacers and the complements of the last letters of the incoming ones. At the j -th visit to x ($j = 0, 1$), let the outgoing spacer begin with $\text{AC}[j]$ and the incoming one end with $\text{CA}[j]$. The letters following the occurrences of P_x are then A, G for a vertex of degree 2 and A, C, G, T for a vertex of degree 4, pairwise different. The interior letters of the spacers are free.

Lemma 3 (realization). *Suppose that in $S \cup \text{rc } S$ the only k -mers occurring more than once are the P_x , the P_x are distinct, and no $P_x \Sigma P_y$ is its own reverse complement. Then the double cover $D_k(S)$ has the following properties:*

1. its vertices are the n palindromes P_x , so $\bar{v} = v$ for every vertex;
2. its contents are the words $P_{x_{i-1}}\Sigma_i P_{x_i}$ and their reverse complements, one ρ -orbit of multiplicity $m = 1$ per edge of G , with no self-complementary content;
3. the quotient of $D_k(S)$ by ρ is G .

Proof. The vertices of D_k are the repeated k -mers [2]; P_x occurs $\deg(x) \geq 2$ times. The contents are the stretches between consecutive vertex occurrences, i.e. the $P\Sigma P$ words of S and their reverse complements in $\text{rc } S$. Each occurs once, so it has multiplicity 1. The orbit of the i -th one joins $P_{x_{i-1}}$ and P_{x_i} , which are fixed by ρ . $\square$

Lemma 4 (the hypotheses can be met in polynomial time). *There are $h, \ell = O(\log m)$ and a deterministic algorithm, polynomial in m , that chooses the words u_x and the spacer interiors so that the hypotheses of Lemma 3 hold.*

Proof sketch. Choose the free letters uniformly at random, with $\ell \geq k$. A *bad event* is one of three things: two occurrences in $S \cup \text{rc } S$ of equal k -mers that are not both occurrences of one P_x ; $P_x = P_y$ for some $x \neq y$; or a self-complementary $P\Sigma P$. Since $\ell \geq k$, a window of length k is an occurrence of some P_x , or it lies inside one spacer, or it straddles one junction between a palindrome and a spacer. In the last two cases it contains at least $h - 2$ free letters from a single source: half of one palindrome, or the interior of one spacer.

Two windows that differ only in fixed letters cannot be equal, by the boundary assignment. This is the reason for it: without it the $(k + 1)$ -mers $P_x \mathbf{c}$ would repeat with probability $1 - 4!/4^4 \approx 0.91$ at a vertex of degree 4. Otherwise, equality of two windows is a system of equations $a_i = b_i$ or $a_i = \bar{b}_i$ between letters. The complement case arises when one window lies in $\text{rc } S$. An equation $a = \bar{a}$ has no solution over $\{\mathbf{A}, \mathbf{C}, \mathbf{G}, \mathbf{T}\}$, and the remaining equations link the free letters in paths and cycles. So the system fixes at least half of them, and its probability is at most $4^{-(h-2)/2}$. There are at most $(2m(k + \ell))^2$ pairs of windows. With $h \geq 2 \log_4(8m^2(k + \ell)^2) + 2$ the expected number of bad events is below $1/2$.

The conditional probability of each bad event, given some letters, is computed exactly from the same system of equations. So the method of conditional expectations fixes the letters one at a time, keeping the expected number of bad events below 1. This is deterministic and polynomial. $\square$

The verification script uses random interiors and checks the hypotheses, retrying on failure. With $k = 12$ and $\ell = 18$ it never needed a retry on the graphs below.

4 The count

Theorem 5. *For S and k as in Section 3,*

$$\#\text{DS}(S, k) = 2\varepsilon(G),$$

and the box points of Bio 8 are in bijection with the Eulerian orientations of G .

Proof. Every orbit has $m = 1$. So a point of the box chooses, for each orbit $\{a, \rho a\}$, exactly one of a and ρa . Since all vertices are palindromic, $a: P_x \rightarrow P_y$ and $\rho a: P_y \rightarrow P_x$ are the two orientations of the edge $\{x, y\}$ of G . A point is a circulation iff the chosen orientation is balanced, i.e. Eulerian. At it, $c(a) \in \{0, 1\}$, so the factorials of $N_{\text{seq}} = \tau \prod (d^+ - 1)! / \prod c(a)!$ disappear and $N_{\text{seq}} = \text{ec}(O)$ [2]. Summing gives Lemma 1. $\square$

graph	n	$ E $	$ S $	$\varepsilon(G)$	Eulerian orientations	$\#DS(S, 12)$
4-regular	3	6	180	16	10	32
4-regular	4	8	240	44	16	88
4-regular	5	10	300	112	28	224
4-regular	6	12	360	328	42	656
4-regular, simple	5	10	300	132	24	264
4-regular, simple	6	12	360	372	38	744
4-regular, simple	7	14	420	1052	60	2104
degrees 2, 4	6	10	300	36	18	72
degrees 2, 4	4	6	180	6	6	12

Table 1: A selection of the 23 graphs of checks (E), (R), (D), (K). $\varepsilon(G)$ is found by exhaustive enumeration of trails. $\#DS$ comes from Bio 8’s lattice sum on the double cover built from the definition, and it equals $2\varepsilon(G)$ in every case. So does the closure of S under Bio 17’s moves. The box points equal the Eulerian orientations in every case.

Corollary 6. *Given a circular DNA sequence S and an even k , computing $\#DS(S, k)$ is $\#P$ -hard, and so is computing the number of points of Bio 8’s box. In particular, no algorithm, closed formula or “meshless” expression evaluates $\#DS$ in time polynomial in $|S|$ unless $FP = \#P$.*

Proof. The map $G \mapsto (S, k)$ is polynomial and $|S| = m(k + \ell) = O(m \log m)$. Divide by 2 and apply Theorem 2. For the box, use [9]. $\square$

Table 1 shows a selection of the verified cases. In every case, the double cover of the constructed genome, built from the definition, has exactly n palindromic vertices, $|E|$ orbits of multiplicity 1 and no self-complementary content. In check (S), 4-regular graphs with $n = 4, \dots, 8$ give genomes of 240–480 bp, linear in $|E|$. On them $\#DS$ grows from 80 to 6 008.

5 Two by-products

Bio 17’s conjecture on this class. Bio 17 conjectured that transpositions, inversions at inverted repeats, and reverse complementation connect $DS(S, k)$ [5]. Here every vertex is its own complement. So an inversion between two visits of P_x (a flip in the sense of [2, §7]) reverses the closed stretch of the circuit between them, which is a κ -transformation. Kotzig proved that the κ -transformations connect all Eulerian circuits of a connected graph [10, 11]. Reversal of the whole circuit supplies the direction.

Corollary 7. *The generation conjecture of Bio 17 holds for every genome of Section 3.*

Check (K) confirms it: the closure of S under inversions and reverse complementation alone is all of $DS(S, k)$ in all 23 cases.

Both counting columns are hard. Bio 12 counted the points of the box (2 481 687 900 for $\phi X174$ at $k = 10$) and Bio 13 the reconstructions. By Theorem 5 the first count is the number of Eulerian orientations of a graph, and the second twice its number of Eulerian circuits. Both are $\#P$ -complete in general.

6 What this does and does not exclude

- *Excluded:* any method polynomial in $|S|$ that is exact on all inputs. That includes a Pfaffian, a determinant, or a Gauss sum with polynomially many polynomially computable terms. Unless $\text{FP} = \#P$, none of them exists. The proposed Gauss-sum expansion $\#DS = |K|^{-1} \sum_{\gamma} \mathcal{G}(\gamma) \Phi(\gamma)$ fails for a second, independent reason. The summation domain is the box, not a group, so Poisson summation over the finite group $K(D_k)$ does not collapse it. Written as a Fourier expansion on K , the $\Phi(\gamma)$ carry the entire difficulty.
- *Not excluded:* algorithms exponential in a structural parameter. Bio 12's frontier elimination is exponential in the frontier width and linear otherwise. Genomes of bounded width are tractable. $\phi X174$ at $k = 10$ has width 26, which is why it was feasible.
- *Not excluded:* efficient approximation. Whether $\#DS$ admits a fully polynomial randomized approximation scheme is open; so is the corresponding question for undirected Eulerian circuits.
- *Odd k .* The construction needs palindromic k -mers, which exist only for even k . For odd k no vertex is its own complement, and the quotient of D_k is the signed graph of Bio 7 [1]. The strand choices are then orientations of a signed graph, and the count is one of Eulerian circuits of a bidirected graph. We expect hardness there too, but do not prove it.

Ancillary file

`bio19_verify.py` (standard library; imports `ds_fermion.py` of Bio 13 and `bio17_verify.py` of Bio 17) runs checks (E), (R), (D), (K) and (S) in two seconds. It writes its transcript to `bio19_output.txt`.

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